Tan Dai Ngo

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EDUCATION

University of Victoria — Master of Engineering, Applied Data Science (Co-op) *Victoria, BC | Sep 2025 - Dec 2027*

- **Coursework:** Algorithms and Data Models, Systems for Massive Datasets. **GPA:** 8.68 / 9.00.

University of Chicago — Master of Science, Computer Science *Chicago, IL, USA | Sep 2020 - Mar 2022*

- **Coursework:** Applied Software Engineering, Distributed Systems, Databases, Networks, Algorithms.

University of Washington — Bachelor of Science, Economics *Seattle, WA, USA | Sep 2015 - Aug 2019*

- **Minors:** Informatics, Applied Mathematics. **GPA:** 3.57 / 4.00.

EXPERIENCE

Software Developer — T-Mobile (via BeaconFire Inc.) *Bellevue, WA, USA | Jun 2022 - Apr 2024*

- Documented the root cause of 20+ defects traced across service boundaries, so each fix outlived the ticket.
- Built and maintained REST endpoints across 5+ RBS (Roaming Business System) services using Java and Spring Boot.
- Deployed features to production through Jenkins across 50+ releases.
- Maintained report response times under 3 seconds on SQL joins spanning multiple source databases.
- Cut manual effort on recurring dataset builds and validation runs by 80% by automating both in Python.

Mathematics and Computer Science Teacher — APU American International Schools *Da Nang City, Vietnam | Aug - Dec 2024*

- Explained technical material to 50+ students with no prior background, adjusting the pitch in real time.

PROJECTS

Multi-Stage IoT Intrusion Detection — Python, scikit-learn, XGBoost, PyTorch *Team Capstone, UVic*

- Compared six detection models under a 1% false-positive budget, reaching 92.5% recall at 0.96% FPR.
- Designed a four-condition experiment to separate genuine learning from leakage in a public benchmark.
- Proved the benchmark encodes capture-session identity, recoverable from flow features at ROC-AUC 0.922.
- Reported the honest cost rather than the headline: PR-AUC falls 0.927 to 0.630 on a session-disjoint split.

Ocean Data Platform with RAG — Python, Spark, XGBoost, Qdrant, FastAPI *Personal Project*

- Harmonized 10M+ sensor observations from NOAA and ONC sources into one layered, queryable model.
- Evaluated retrieval over dataset metadata against a held-out query set: hit@k 1.0, term recall 0.875.
- Forecast sensor readings with XGBoost on chronological holdouts rather than random splits.
- Containerized every service so each experiment could be reproduced from scratch by someone else.

Distributed Facility Reservation System — Python, FastAPI, SQLite, pytest *Team Project, UChicago*

- Designed 27 versioned REST endpoints and published OpenAPI documentation for every one of them.
- Wrote 71 pytest tests across the API, database, and reservation-rules layers, run on each change.
- Negotiated one booking contract with four teams, then tested against their independently built services.
- Secured every endpoint behind session tokens checked for freshness and permission scope.

Anomaly Detection at Scale on Species Data — Python, PySpark *Team Project, UVic*

- Implemented Isolation Forest from the published algorithm rather than a library, then validated the result.
- Identified a previously undescribed variant of feature-specific swamping and proposed two mitigations.
- Verified linear runtime growth on 1,093,203 records using stratified samples across eight quantile bins.

TECHNICAL SKILLS

Languages: Python, Java, SQL
AI Tooling: Claude Code for test authoring, debugging, refactoring, documentation, and plan validation
Web and APIs: REST API design, FastAPI, Spring Boot, OpenAPI documentation, session-based auth
Data and ML: pandas, NumPy, scikit-learn, XGBoost, PyTorch, Spark, PySpark, Kafka, Cassandra, Qdrant vector store
Engineering Practice: Git, GitHub pull requests, code review, pytest, Docker, Jenkins, relational schema design